



Methods for Calculating Levodopa Equivalent Daily Dose (LEDD) in PPMI Data

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Summary

PPMI collects the dosage of Parkinson's Disease (PD) medications taken by study participants on the LEDD log (LEDDLOG), which can be converted to a Levodopa Equivalent Daily Dose (LEDD) using the methods described in Tomlinson, et al and updated by Schade, et al and Jost, et al.

Methods

A column titled "LEDD" is recorded in LEDDLOG table to show the LEDD for each individual PD medication listed in the log. The LEDD is calculated by first using an algorithm to calculate the daily dose (mg) of each PD medication. The medication dose and frequency are taken from the LEDDLOG table. Next, the daily dose for each PD medication is multiplied by the appropriate conversion factor listed in Table 1. These conversion factors were calculated using the methods described in Tomlinson, et al; Schade, et al; and Jost, et al.

For PD medications except COMT inhibitors, the column LEDD shows the value of the Levodopa equivalent dose for that medication. For COMT inhibitors, the column LEDD reads "LD x 0.33" (for Entacapone) or "LD x 0.5" (for Opicapone/Tolcapone). To find the LEDD for COMT inhibitors, first find the subtotal LEDD of Levodopa only, and then multiply the Levodopa LEDD by either 0.33 or 0.5 as instructed.

Safinamide, Zonisamide, and Trihexyphenidyl all give a single daily LEDD, regardless of dose.

To find the total LEDD at a specific time point, add all values of the column LEDD for the PD medications taken at that time point. Other PD medications that are not included in the calculation of the total LEDD will have a missing value for the LEDD column.

LEDD calculations are updated periodically when new information becomes available.





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Table 1. Conversion factors used to calculate LEDDs for PD drugs

Drug Class	Drug	Conversion factor
Dopamine Replacement	Levodopa	DD x 1
	Dual-release levodopa (Madopar DR)	DD x 0.85
	Controlled-release levodopa	DD x 0.75
	Extended-release levodopa (Rytary)	DD x 0.5
	Intrajejunal levodopa/carbidopa infusion	DD x 1.11
	Intrajejunal levodopa/carbidopa/entacapone infusion	DD x 1.11 (morning) + DD x 1.46 (maintenance and extra doses)
	Subcutaneous foslevodopa/foscarbidopa	DD x 0.75
	Inhaled levodopa (Inbrija)	DD x 0.69 (capsules)
COMT Inhibitors*	Entacapone	LD x 0.33
	Tolcapone	LD x 0.5
	Opicapone	LD x 0.5
Dopamine Agonists	Pramipexole, Lisuride, Pergolide	DD x 100
	Ropinirole	DD x 20
	Rotigotine	DD x 30.3
	Piribedil	DD x 1
	Apomorphine injected	DD x 10
	Apomorphine sublingual (Kynmobi)	DD x 1.5
	Bromocriptine	DD x 10
	Cabergoline	DD x 66.7
	Dihydroergocryptine (DHEC)	DD x 5
MAOB Inhibitors	Selegiline oral	DD x 10
	Selegiline sublingual	DD x 80
	Rasagiline	DD x 100
Other	Amantadine	DD x 1
	Amantadine ER (Gocovri)	DD x 1.25
	Amantadine ER (Osmolex)	DD x 1
	Safinamide (Xadago)	LED = 150
	Zonisamide	LED = 100
	Trihexyphenidyl	LED = 100
	Istradefylline (Nourianz)**	LD x 0.2
	Mucuna pruriens	DD x 0.013

DD = daily dose of drug being converted; LD = levodopa dose; LED = levodopa equivalent dose.





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* Regardless of the COMT inhibitor dose, the subtotal LED of levodopa-containing medications is multiplied by the COMT conversion factor to give the LED of the COMT inhibitor.

** Regardless of the Istradefylline dose, the subtotal LED of levodopa-containing medications and COMT inhibitors is multiplied by 0.2 to give the LED of Istradefylline.

The following PD drugs are NOT included in the LEDD calculation: Propranolol; the following anti-cholinergics: Benztropine/Cogentin, Biperiden/Akineton, Budipin/Parkinsan; Investigational drugs.

References

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